Hypergeometric Distribution

1 Definition

A discrete random variable X is said to have a Hypergeometric Distribution if it's PMF is given by,

$$f_X(x) = \left\{ \begin{array}{cc} \frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}}, & x = 0, \dots n; 0 < p < 1 \\ 0, & \text{otherwise} \end{array} \right\}$$

Note that,

$$\begin{aligned} & 0 \leqslant x \leqslant Np, \quad 0 \leqslant n - x \leqslant Nq \\ \Rightarrow\ & n - Nq \leqslant x \leqslant n \\ \Rightarrow\ & \max\{0, n - Nq\} \leqslant x \leqslant \min\{Np, n\} \end{aligned}$$

We know,

$$\begin{aligned} & \sum_x \binom{M}{x}\binom{N-M}{n-x} = \binom{N}{n} \\ \Rightarrow\ & \frac{\sum_x \binom{M}{x}\binom{N-M}{n-x}}{\binom{N}{n}} = 1 \quad \text{(this term will form a pmf)} \\ \therefore\ & \sum_x \frac{\binom{Np}{x}\binom{N-Np}{n-x}}{\binom{N}{n}} = \sum_x \frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} = 1 \end{aligned}$$

2 Moments

First, we will focus on factorial moments.

$$\begin{aligned} E\left(X_r\right) &= \sum_{x=0}^{n} x_{(r)} \frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} \\ &= \sum_{x=0}^{n} x_{(r)} \frac{\frac{(Np)!}{x!(Np-x)!}\binom{Nq}{n-x}}{\binom{N}{n}} \\ &= \frac{(Np)_r}{(x)_r} \sum_{x=r}^{n} \frac{\frac{(Np-r)!}{(x-r)!(Np-x)!}\binom{Nq}{n-x}}{\binom{N}{n}} \\ &= \frac{(Np)_r}{\binom{N}{n}} \sum_{x=r}^{n} \binom{Np-r}{x-r}\binom{Nq}{x-r} \\ &= \frac{(Np)_r}{\binom{N}{n}} \sum_{y=x-r=0}^{n-r} \binom{Np-r}{y}\binom{Nq}{n-r-y} \\ &= \frac{(Np)_r}{\binom{N}{n}}\binom{N-r}{n-r} \underbrace{\sum_{y=0}^{n-r} \frac{\binom{Np-r}{y}\binom{Nq}{n-r-y}}{\binom{N-r}{n-r}}}_{\text{PMF of Hypergeometric}} \end{aligned}$$

$$= \frac{(Np)_r}{\binom{N}{n}} \binom{N-r}{n-r}$$

$$E(X_r) = \frac{(Np)_r}{\binom{N}{n}} \binom{N-r}{n-r}$$

$r = 1, E(X) = \frac{Np}{\binom{N}{n}} \binom{N-1}{n-1} = np$

$r = 2, E(X_2) = E(X(X-1)) = \frac{p(Np-1)n(n-1)}{(N-1)}$

$$\begin{aligned}
\text{Var}(X) &= E\left(X^2\right) - E^2(X) \\
&= E(X(X-1)) + E(X) - E^2(X) \\
&= \frac{p\,(Np-1)\,n(n-1)}{(N-1)} + np - n^2p^2 \\
&= \frac{p(Np-1)n(n-1) + np(N-1) - n^2p^2(N-1)}{(N-1)} \\
&= \frac{np}{(N-1)}[(Np-1)(n-1) + (N-1) - np(N-1)] \\
&= \frac{np}{(N-1)}\,[Npn - n - Np + 1 + N - 1 - npN + np] \\
&= \frac{np}{(N-1)}[(N-n) - p(N-n)] \\
&= \frac{np}{(N-1)}(N-n)(1-p) \\
&= npq\frac{(N-n)}{(N-1)}
\end{aligned}$$

1. Suppose there are N individuals in a population of which Np individuals possess a character A and Nq do not possess the character A. Let X be the number of individuals with characteristics A in a random sample of size n. Find $P(X = x)$ under the following scheme WR and WOR.

i) **WR (With Replacement)**

X denote the no of individual in the sample who possess the character A.

$P(X = x) = P(x$ no of individuals with character A and (n-x) with characterA^c in a random sample of size $n)$

$$\begin{aligned}
&= \binom{n}{x}\left(\frac{Np}{N}\right)^x\left(\frac{Nq}{N}\right)^{n-x} \\
&= \frac{\binom{n}{x}(Np)^x(Nq)^{n-x}}{N^n} \\
&= \binom{n}{x}p^xq^{n-x}.
\end{aligned}$$

$$X_{\text{WR}} \sim \text{Bin}(n, p)$$

ii) **WOR (Without Replacement)**

$$P(X = x) = \frac{\binom{n}{x}(Np)_x(Nq)_{n-x}}{\binom{N}{n}} \quad \text{(considering ordered sample)}$$

$$= \frac{\binom{Np}{x}\binom{Na}{n-x}}{\binom{N}{n}}$$

$$X_{\text{WOR}} \sim \text{Hyp}(n, N, p)$$

2. Show that, $\frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} \xrightarrow{N\to\infty} \binom{n}{x}p^x q^{n-x}$

As, $N \to \infty$, $\quad \frac{n}{N} \to 0$.

$$\begin{aligned}
\frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} &= \binom{n}{x}\frac{(Np)_x(Nq)n-x}{(N)_n} \\
&= \binom{n}{x}\frac{Np(Np-1)\dots(Np-x+1)Nq(Nq-1)\dots(Nq-n+x)}{N(N-1)(N-2)\dots(N-n+1)} \\
&= \binom{n}{x}\frac{p\left(p-\frac{1}{N}\right)\left(p-\frac{2}{N}\right)\dots\left(p-\frac{x-1}{N}\right)q\left(q-\frac{1}{N}\right)\dots\left(q-\frac{n-x-1}{N}\right)}{1\left(1-\frac{1}{N}\right)\left(1-\frac{2}{N}\right)\dots\left(1-\frac{n-1}{N}\right)} \\
&= \binom{n}{n}p^x q^{n-x}
\end{aligned}$$

Interpretation

As N tends to infinity, finite population become infinite population. From an infinite population drawing sample under WOR is practically equivalent to draw sample under WR. As a sample under WR can also be obtained under WOR. Hence, as N tends to infinity, the distribution of WOR becomes the distribution of WR, Hypergeometric distribution becomes Binomial distribution.

Remark

$$\begin{aligned}
&E\left(X_{\mathrm{WR}}\right) = E\left(X_{\mathrm{WOR}}\right) = np. \\
&\mathrm{Var}\left(X_{\mathrm{WR}}\right) = npq > npq\underbrace{\left(\frac{N-n}{N-1}\right)}_{<1} = \mathrm{Var}\left(X_{\mathrm{WOR}}\right).
\end{aligned}$$

The factor in the variance $\left(\frac{N-n}{N-1}\right)$ as a multiplier using due to the use of WOR schemes in a finite population instead of an infinite population. It is known as **Finite population correction (FPC).**

3 Examples:

1. An unknown number of animals inhabit in a certain region. First, catch a number of m of these animals, mark them in some manner and release them. After allowing the marked animals time to dispersed through at the region. A new catch of size n is made. Let X denotes the number of marked animals in the second capture. Find the probability distribution of X. Also, find $E(X)$ and Var(X).

Let, there are total N number of unknown animals. Out of these m are marked, so, $(N-m)$ are unmarked.

$$\begin{aligned}
P(X=x) &= \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}, x = 0(1)n. \\
\therefore X &\sim \text{ Hypergeometraic } (m, N, p) \\
E(X) &= \frac{nm}{N}, \frac{m}{N} = p \\
V(X) &= npq\left(\frac{N-n}{N-1}\right) = n\frac{m}{N}\left(1-\frac{m}{N}\right)\left(\frac{N-n}{N-1}\right)
\end{aligned}$$

2. If $X_{\text{Hyp}}(n, N, p)$, show that $E(n - X)_r = (n)_r \frac{(Nq)_r}{(N)_r}$

$$
\begin{aligned}
X &\sim \text{Hyp}(n, N, p) \\
Y = n - X &\sim \text{Hyp}(n, N, 1 - p = q) \\
P(Y = y) = P(n - X = y) &= P(X = n - y) \\
&= \frac{\binom{Np}{n-y}\binom{Nq}{y}}{\binom{N}{n}} \\
&= \frac{\binom{Nq}{y}\binom{Np}{n-y}}{\binom{N}{n}} \sim \text{Hyp}(n, N, q)
\end{aligned}
$$

We got, $E(X_r) = \frac{(Np)_r \binom{N-r}{n-r}}{\binom{N}{n}}$

$$
\begin{aligned}
E(Y_r) = E[(n - X)_r] &= \frac{(Nq)_r \binom{N-r}{n-r}}{\binom{N}{n}} \\
&= (Nq)_r \frac{(N-r)!}{(n-r)!(N-n)!} \frac{n!(N-n)!}{N!} \\
&= (Nq)_r \frac{(n)_r}{(N)_r}
\end{aligned}
$$

3. $X \sim \text{Hyp}(n, N, m)$, show that $E\left(X^k\right) = \frac{nm}{N} E(Y + 1)^{k-1}$, where, $Y \sim \text{Hyp}(n - 1, N - 1, m - 1)$. Find $E(X)$, $\text{Var}(X)$.

$$
\begin{aligned}
E\left(X^k\right) &= \sum_{x=0}^{n} x^k \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}} \\
&= \sum_{x=0}^{n} x^{k-1} \frac{\binom{m-1}{x-1} m \binom{N-m}{n-x}}{\binom{N}{n}} \\
&= \sum_{x=1}^{n} \frac{x^{k-1} m \binom{m-1}{x-1} \binom{(N-1)-(m-1)}{(n-1)-(x-1)}}{\frac{N}{n}\binom{N-1}{n-1}} \\
&= \sum_{x-1=y=0}^{n} \frac{(y+1)^{k-1}\binom{m-1}{y}\binom{N-1-y}{n-1-y}}{\binom{N-1}{n-1}} \frac{nm}{N} \\
&= \frac{nm}{N} E(Y + 1)^{k-1}; \quad \text{as } Y \sim \text{Hyp}(n - 1, N - 1, m - 1)
\end{aligned}
$$

$$
\begin{aligned}
K = 1, E(X) &= \frac{nm}{N} E(Y + 1)^0 = \frac{nm}{N} \\
K = 2, E\left(X^2\right) &= \frac{nm}{N} E(Y + 1) \\
&= \frac{nm}{N}\left[\frac{(n-1)(m-1)}{(N-1)} + 1\right] \\
&= \frac{nm}{N}\left(\frac{nm - m - n}{N - 1}\right)
\end{aligned}
$$

$$
\begin{aligned}
V(X) &= E\left(X^2\right) - E^2(X) \\
&= \frac{nm}{N}\left(\frac{nm - m - n}{N - 1}\right) - \frac{n^2 m^2}{N^2} \\
&= \frac{nm}{N} \frac{(N - m)(N - n)}{N(N - 1)}
\end{aligned}
$$

4. Let $X_1, \dots X_n$ be n independent Bernoulli(p) RV. Let $S_j = \sum_{i=1}^{J} X_i$ where $J = 1(1)n$. Show that, $P(S_m = r \mid S_n = s)$ does not depend on p, where $m < n$. Identify the distribution and obtain first two moments of it.

$$X_1, X_2 \dots X_n \overset{\text{iid}}{\sim} \text{Bern}(p)$$

$$S_j = \sum_{i=1}^{J} X_i \sim \text{Bin}(J, p), \quad J = 1(1)n.$$

$$S_m = \sum_{i=1}^{m} X_i \sim \text{Bin}(m, p).$$

$$\begin{cases} S_n = X_1 + \cdots + X_m + X_{m+1} + \cdots + X_n \\ S_m = X_1 + \cdots + X_m; \quad n > m \end{cases}$$

$$S_n = \sum_{i=1}^{n} X_i \sim \text{Bin}(n, p)$$

$$S_n - S_m = \sum_{i=m+1}^{n} X_i \sim \text{Bin}(n-m, p)$$

$$P(S_m = r \mid S_n = s) = \frac{P(S_m = r, S_n = s)}{P(S_n = s)}$$

$$= \frac{P(S_m = r, S_n - S_m = s - r)}{P(S_n = s)}$$

$$= \frac{P(S_m = r)\, P(S_n - S_m = s - r)}{P(S_n = s)}$$

$$= \frac{\binom{m}{r} p^r q^{m-r} \binom{n-m}{s-r} p^{s-r} q^{n-m-s+r}}{\binom{n}{s} p^s q^{n-s}}$$

$$= \frac{\binom{m}{r}\binom{n-m}{s-n}}{\binom{n}{s}}, r = 0(1)s.$$

$$E(X) = \frac{sm}{n}$$

$$\text{Var}(X) = s\frac{m}{n}\left(1 - \frac{m}{n}\right)\left(\frac{n-s}{n-1}\right)$$

4 Mode of Hypergeometric Distribution

Note that, $\dfrac{f(x)}{f(x-1)} = \dfrac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} \dfrac{\binom{N}{n}}{\binom{Np}{x-1}\binom{Nq}{n-x+1}}$

$$\frac{f(x)}{f(x-1)} = \frac{\binom{Np}{x}\binom{Nq}{n-x}}{\binom{N}{n}} \frac{\binom{N}{n}}{\binom{Np}{x-1}\binom{Nq}{n-x+1}}$$

$$= \frac{(Np - x + 1)(n - x + 1)}{(Nq - n + x)x} \gtreqless 1 \text{ if}$$

$$x \gtreqless \frac{(n+1)(Np+1)}{(N+2)}$$

Case I

$\dfrac{(n+1)(mp+1)}{(n+2)}$ is not an integer. Let $\left[\frac{(n+1)(NP+1)}{(N+2)}\right] = k$.

$$\frac{f(1)}{f(0)} > 1, \frac{f(2)}{f(1)} > 1, \ldots, \frac{f(k)}{f(k-1)} > 1, \ldots \frac{f(k+1)}{f(k)} < 1, \ldots$$
$$\Rightarrow f(0) < f(1) < f(2) < \ldots < f(k-1) < f(k) > f(k+1) > \ldots$$

Clearly, $f(k)$ is the maximum,

$$\therefore k = \left[\frac{(n+1)(Np+1)}{(N+2)}\right] \text{ is the mode.}$$

Case II

$$k = \frac{(n+1)(Np+1)}{(k+2)} \text{ is an integer.}$$
$$\frac{f(1)}{f(0)} > 1, \frac{f(2)}{f(1)} > 1, \ldots, \frac{f(k)}{f(k-1)} = 1, \frac{f(k+1)}{f(k)} < 1, \ldots$$

$$f(0) < f(1) < f(2) \ldots < f(k-1) = f(k) > f(k+1) > f(k+2) > \ldots$$

Clearly, $f(k-1)$ and $f(k)$ is the maximum, hence, $k\&(k-1) = \frac{(n+1)(Np+1)}{(N+2)}$ are the modes.